
Domestic Service Robot

Heating Dynamics

in PDDL+

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Problem Overview: The Kitchen Environment



Robotic Arm



Water Dispenser



Heater Base



Kettle



Cups



Ingredients

AGENT

1 Single-arm Robot

APPLIANCES

Dispenser (Water source)
Heater base (Heat source)

CONTAINERS

Kettle (Heating)
Cups (Target)

CONSTRAINT

Single-arm kinematics: cannot
manipulate multiple items
simultaneously.

The Core Challenge: Thermal Constraints

FLAVORED WATER

$$T \leq 30^{\circ}\text{C}$$

WARM TEA

$$55 < T \leq 65^{\circ}\text{C}$$

HOT INFUSION

$$85 < T \leq 95^{\circ}\text{C}$$

Goal: Serve correctly prepared beverages.

The Difficulty: Temperatures are not assigned directly; they evolve continuously over time based on active and passive thermodynamics.

The Intuition: The planner must coordinate **spatial actions** (moving objects) with **continuous temporal processes** (waiting for heating/cooling).

State Representation & Mechanics



Continuous Numeric Fluents

- Physics mapped via real-valued variables (temperature, water-volume).
- Replaces complex boolean state-space.



Discrete Action Space

- Spatial manipulation (pick, place, pour).
- Single-arm kinematics strictly managed via mutually exclusive **handempty** and **holding** logic

THE MECHANICAL BOTTLENECK

Goal: Prevents heuristic action-stacking (infinite moves at $t=0$).

Inertia: Robot must wait for the timer to clear before the next action.

```
;; Action triggers inertia
:precondition (<= (robot-cooldown) 0.0)
:effect (assign (robot-cooldown) 1.0)

;; Continuous process drains timer
(:process robot-cooling-down
 :precondition (> (robot-cooldown) 0.0)
 :effect (decrease (robot-cooldown) (* #t
 1.0)))
```

Continuous Physics & Environmental Autonomy



Simultaneous Thermodynamics

Conflicting continuous effects resolved natively by ENHSP solver, avoiding manual state-space calculations.

```
(:process heating
:precondition (heater-on ?h)
:effect (increase (temperature ?k) (* #t 2.5))
(:process cooling-kettle
:effect (decrease (temperature ?k) (* #t 0.5))
```

- Forces the solver to reason over continuous thermal trajectories rather than discrete states.



Autonomous Safety (Events)

The environment reacts independently to physical boundaries, breaking the passive world assumption.

```
TEMPERATURE > 100°C
    ↓
EVENT: OVERHEAT_SAFETY
    ↓
EFFECT: (NOT (HEATER-ON))
```

- Acts as a hard physical limit. Overrides planner actions to prevent hardware failure.

State Transfer & The Logic Gates



Fluid & Thermal Transfer

Action: pour-water

Acts as a state-transfer mechanism, dynamically propagating continuous physical properties across objects.

```
(:action pour-water
:effect (and
  (not (cup-empty ?c))
  (decrease (water-volume ?k) 250.0)
  (assign (temperature-cup ?c) (temperature ?k)))
```

- The target container inherits the continuous thermal state, initiating its own dissipation.



The Logic Gates (Preparation)

Preparation actions are gated by strict mathematical boundaries, rather than just logical states.

```
(:action prepare-warm-tea
:precondition (and
  (holding-ingredient ?i)
  (> (temperature-cup ?c) 55.0)
  (<= (temperature-cup ?c) 65.0))
:effect (beverage-ready-warm-tea ?c))
```

- The planner is mathematically forced to wait for passive cooling to enter the thermal window.